

Course Code : ECT 353

ISJT/MW – 23 / 1807

**Fifth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

MICROCONTROLLERS AND INTERFACING

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions carry marks as indicated.
 - (2) Assume suitable data wherever necessary.
-
1. Recall at least five various Bit handling instructions with examples and explain them in detail. 7(CO1)
 2. Infer with diagram and explain pin and port configurations how an 8051 microcontroller can access a 4 KB external RAM. 7(CO2)
 3. Implement a display circuit comprising of a single 7 Segment Common Anode interfaced with 8051 at port 1. Write a program to display 0 to F continuously with a delay of 1 second. Show the look up table as well as the delay calculations. 8(CO3)
 4. Organize the circuit of 8051 in such a way that serial communication can take place between 8051 and a IBM computer. Write a program of 8051 to transmit string "hello world" at 9600 bps. 8(CO3)
 5. Recognize and recall ARM Modes of operations in detail. 7(CO4)
 6. Differentiate the interrupt handling capacity of ARM7TDMI over 8051 microcontroller. 7(CO4)

ISJT/MW-23 / 1807

Contd.

7. Evaluate a 32 bit multiplication program of ARM7TDMI and explain in detail how data is brought and stored in regards to register set available with ARM7TDMI.

8(CO1)

8. Create an round robin scheduling algorithm for 4 tasks considering the following input data where the time slice to be considered is of 2 time slots each.

TASK NO	Arrival Time	Execution time
1	3	6
2	7	3
3	12	7
4	13	12

8(CO3,4)

